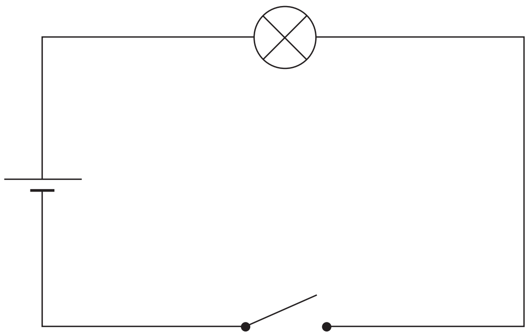
Examiner only

Answer **all** *the questions in the spaces provided.*

1. Students are investigating how the current through a lamp varies with the voltage across it.



(a) (i) Part of the circuit used is shown above. **Add** a variable resistor, ammeter and voltmeter to the circuit diagram. [3]

(ii) Describe the method you would use to take a range of results. [3]

....

....

....

....

(b) Results from the experiment are shown below.

Voltage (V)	Current (A)
0.0	0.0
0.5	0.5
2.0	0.8
4.0	1.1
6.0	1.4
8.0	1.6
10.0	1.8
12.0	1.9

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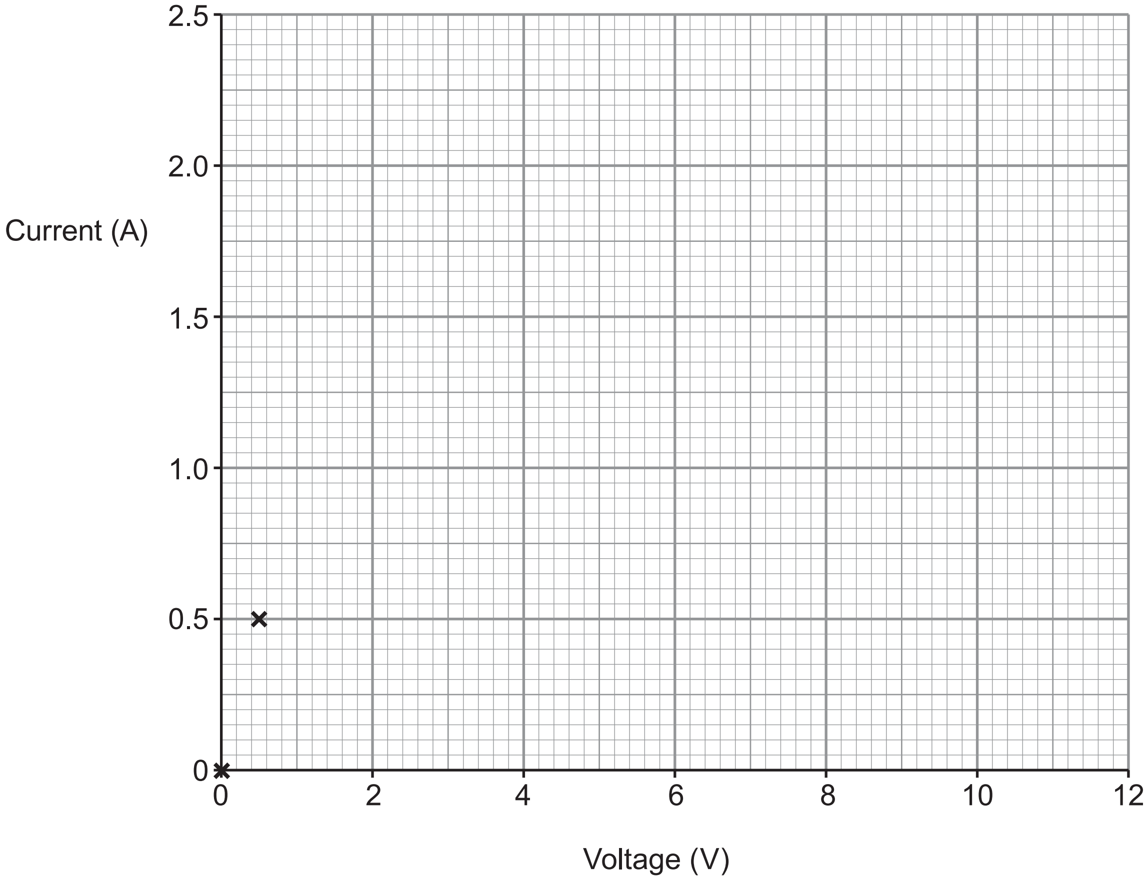
(i) Plot the data on the grid below and draw a suitable line. [3]

The first two points have been plotted for you.



(ii) Describe the relationship between current and voltage for the lamp. [2]

....

....

....

(c) (i) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp at 0.5 V. [2]

resistance = ... Ω

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(ii) State how the resistance of the lamp varies between 0.5 V and 12 V. [1]

....

(d) (i) Use the equation:

power = voltage × current

to calculate the power of the lamp at 0.5 V. [2]

power = ... W

(ii) State how the power of the lamp varies between 0.5 V and 12 V. [1]

....

(e) The experiment was repeated with a length of wire at constant temperature. The current through the wire was 1.5 A when the voltage across it was 12 V.

Add a line to your graph to show how current varies with voltage for this wire. [2]

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